

IT332/CS446 Cloud Computing

2 Understanding AWS

Computing Services - EC2

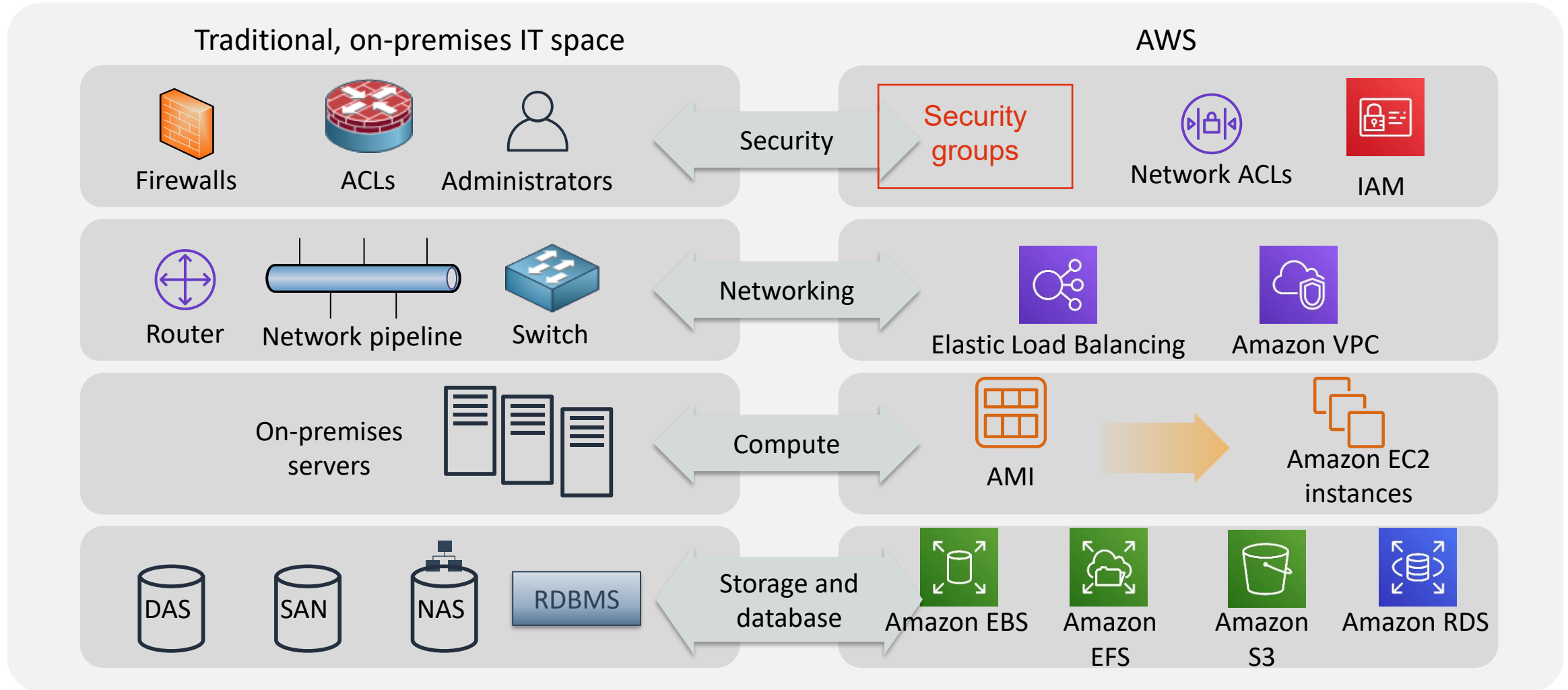
Information Technology
School of Information Technology and Innovation
Bangkok University

Joe Lersak Limwiwatkul

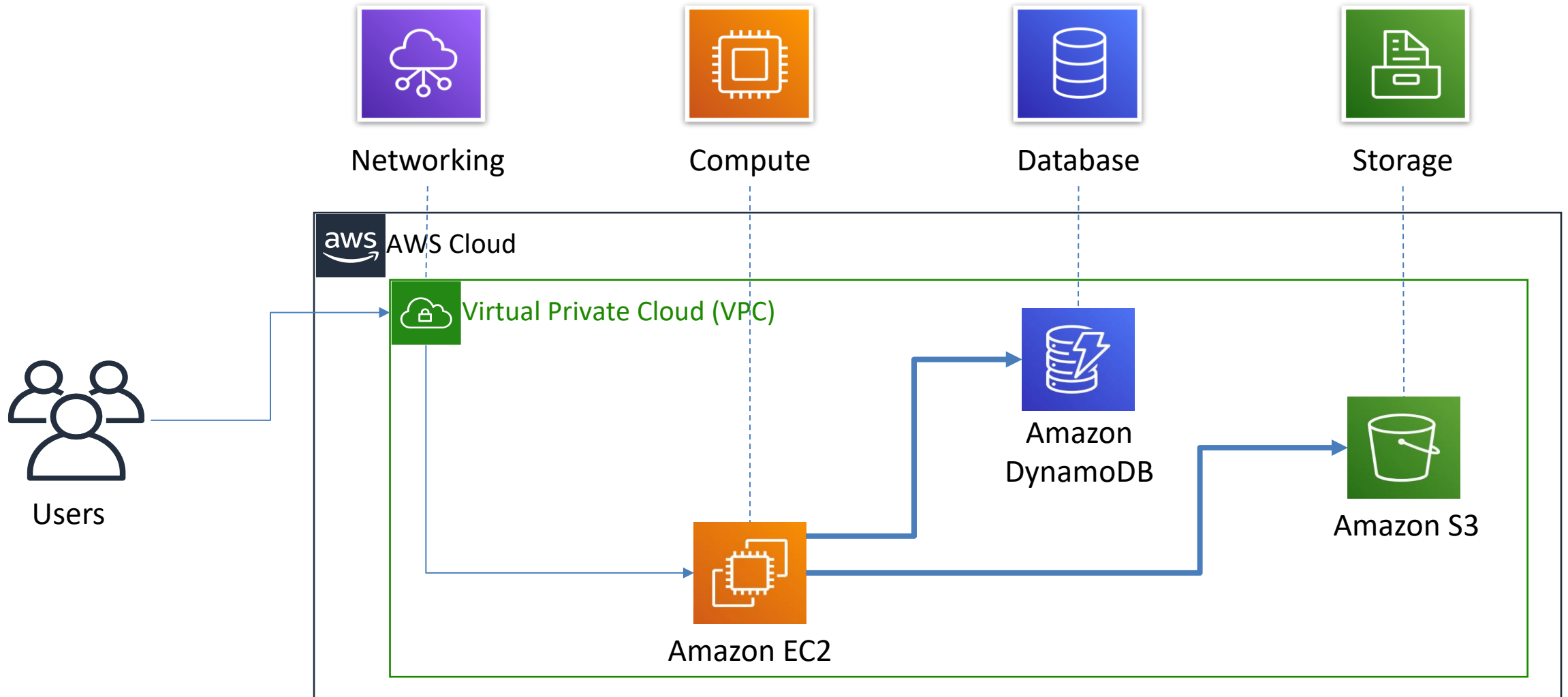
CISSP, CISM, CISA, GSSP-JAVA, ITILv3 Expert, IRCA-ISMS, IRCA-BCMS, RHCE



Migrate to AWS Cloud Computing

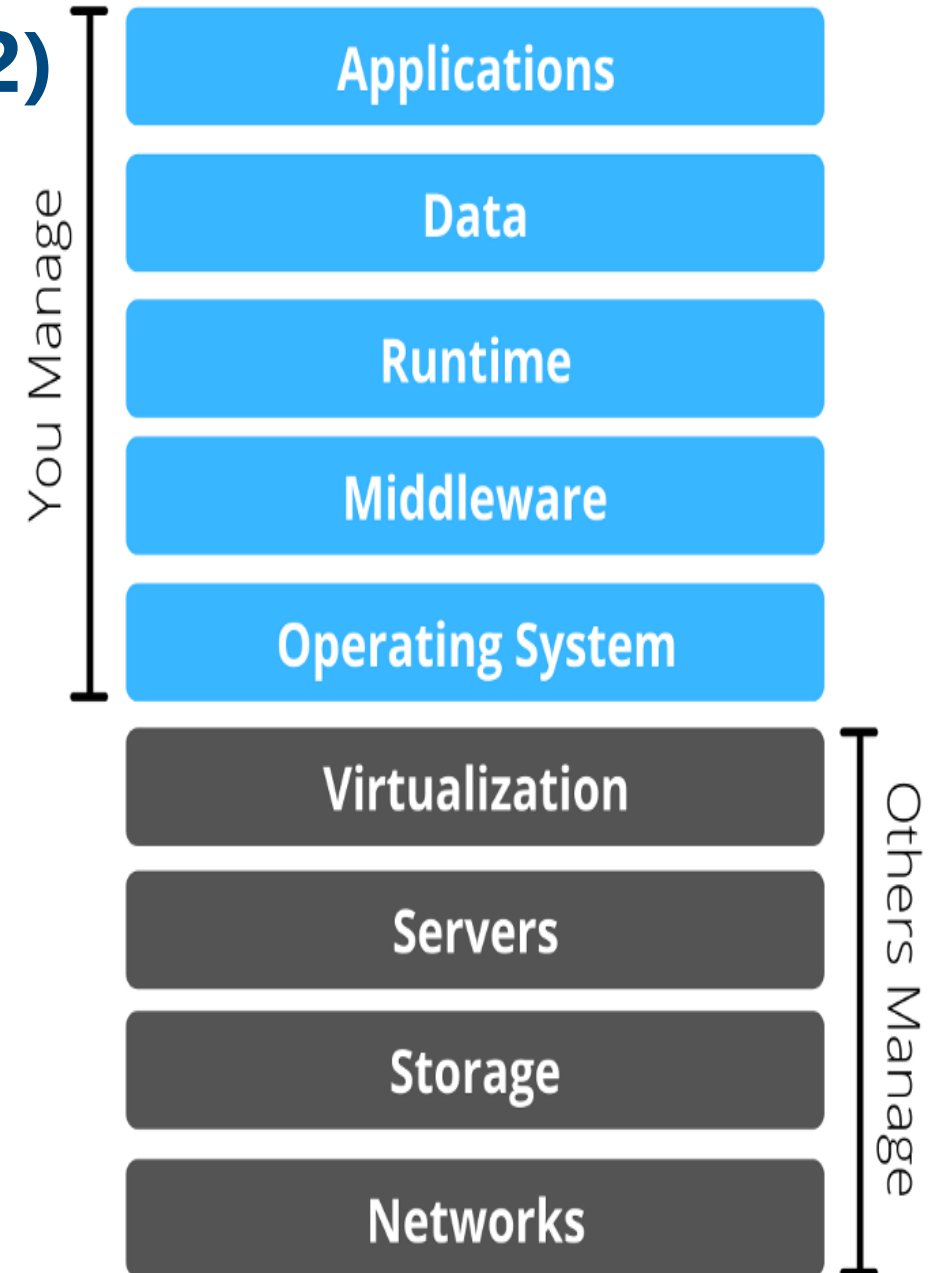


Simple AWS Cloud Computing

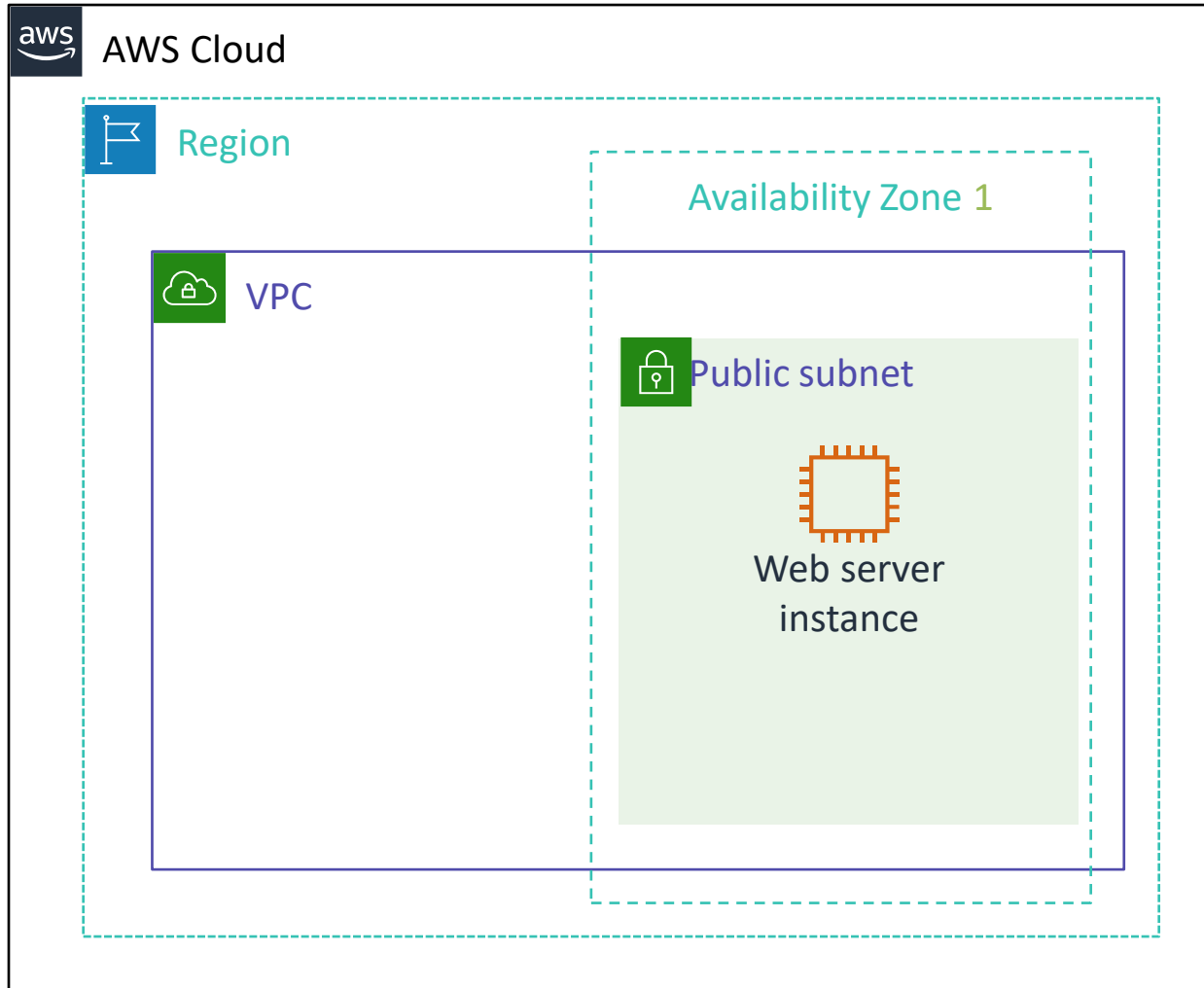


Amazon Elastic Compute Cloud (EC2)

- On-demand, scalable computing resources
 - Provides [virtual machines](#)—referred to as EC2 instances—in the cloud.
 - Gives you full control over the guest operating system (Windows or Linux) on each instance.
- Launch instances of any size into an Availability Zone anywhere in the world.
 - Launch instances from [Amazon Machine Images](#) (AMIs).
 - Launch instances with a few clicks or a line of code, and they are ready in minutes.
 - You can control traffic to and from instances



Understanding Computing Services Lab



- Create our Instance
 - In Lab Network
- Manage Security Rule
- Access to our Machine
- Setup Web Server

A silver golf cart is parked on a sandy beach. The cart is facing left and has a tan roof. In the background, the ocean stretches to the horizon under a clear blue sky with a few wispy clouds. To the right of the cart, there is a blue and white trash can. The text "Access AWS Lab" is overlaid in white on the lower left side of the image.

Access AWS Lab

Access AWS Lab

- Search: [aws academy student login](#)

aws academy student login

- Look for Untitled



Instructure

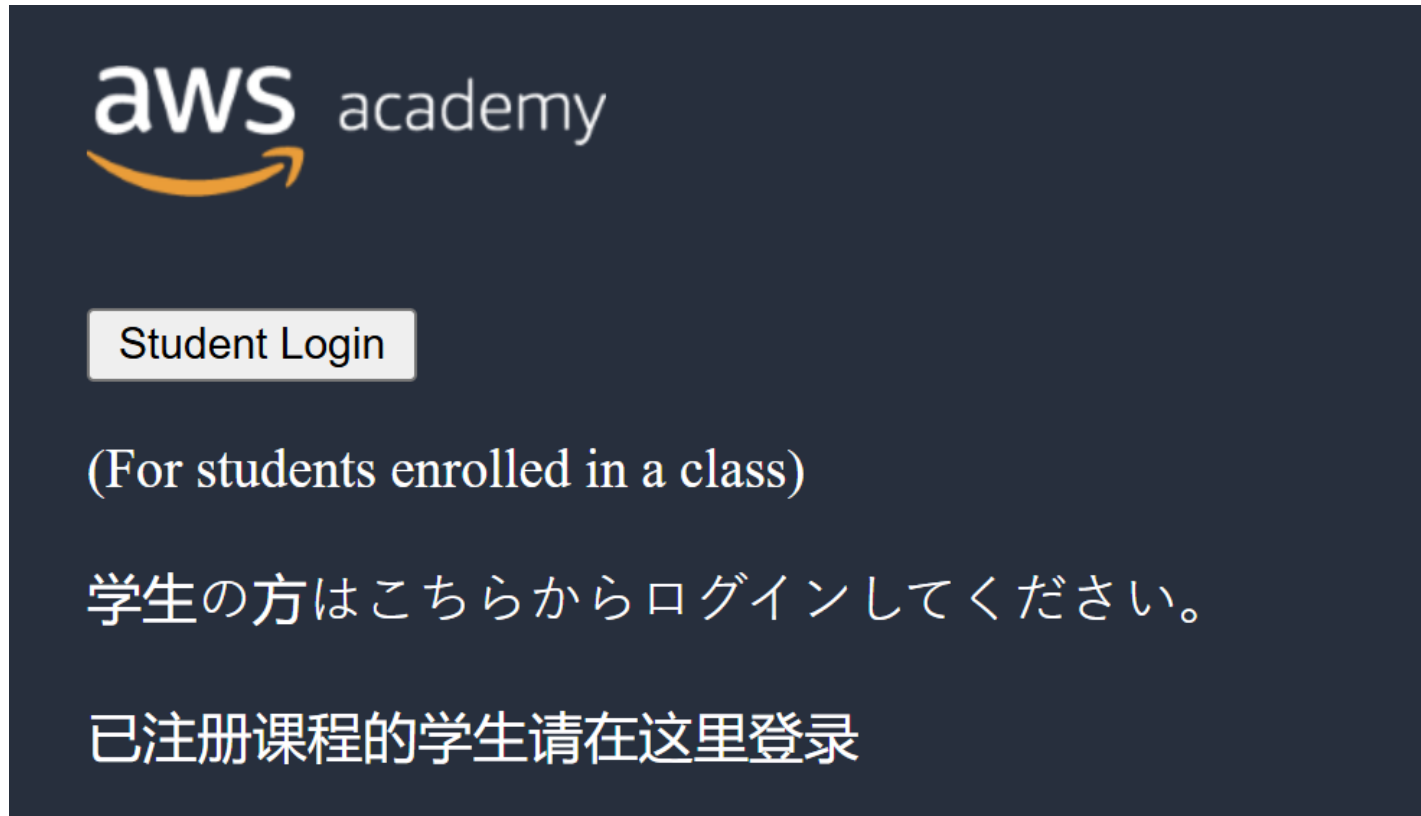
<https://awsacademy.instructure.com> · [Translate this page](#) ⋮

Untitled

学生の方はこちらからログインしてください。已注册课程的学生请在这里登录. **Educator Login.** (For educators who have **access** to the **AWS Academy Portal**). [Read more](#)

Access AWS Lab

- Choose Student Login



Access AWS Lab

- Login with bumail.net
- OTP in your email



Welcome to Canvas

Email *

Password *

☐ Remember me

[Log In](#)

[Forgot password?](#)

[Trouble logging in?](#)

Access AWS Lab

[AWS Academy Cloud Foundations \[180837\]](#)

[AWS Academy Learner Lab \[180838\]](#)

- AWS Academy Learner Lab
 - Use for AWS Lab Hands-On
- AWS Academy Cloud Foundation
 - Prepare for AWS Cloud Practitioner Certification Exam

AWS Academy Cloud Foundations

- Self Study for AWS Academy Cloud Foundations in AWS Academy

Home	▶ Module 1 - Cloud Concepts Overview
Announcements	
Pages	
Modules	▶ Module 2 - Cloud Economics and Billing
Discussions	
Grades	
Lucid (Whiteboard)	▶ Module 3 - AWS Global Infrastructure Overview
	▶ Module 4 - AWS Cloud Security
	▶ Module 5 - Networking and Content Delivery
	▶ Module 6 - Compute
	▶ Module 7 - Storage
	▶ Module 8 - Databases

Access AWS Lab

- First page
 - Indicating AWS Academy Learner Lab
 - Ignore number in the bracket

 ALLv2EN-US-LTI13-134888

Home

Pages

Modules

Discussions

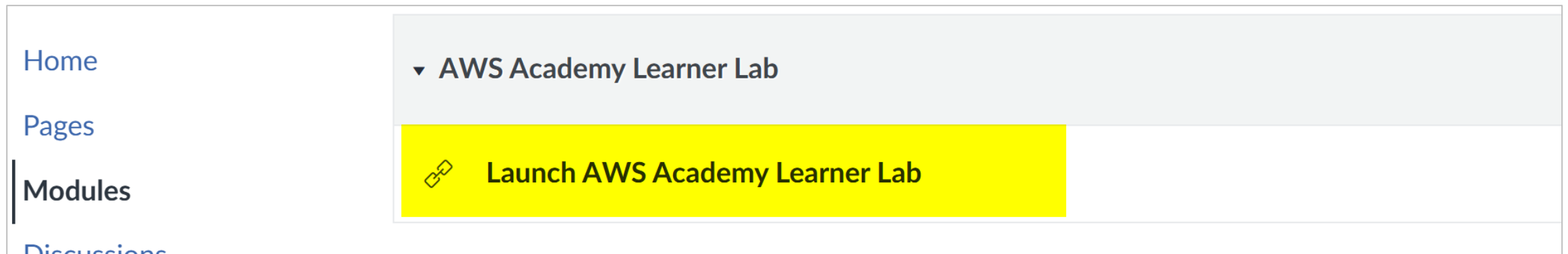
Grades

AWS Academy Learner Lab [134888]

[AWS Academy Learner Lab](#)

Access AWS Lab

- Click **Modules** Tab on the left panel
- Click **Launch AWS Academy Learner Lab**

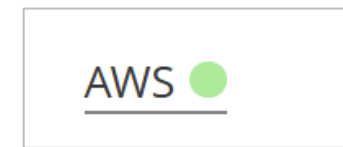


Access AWS Lab

- Click **Start Lab**

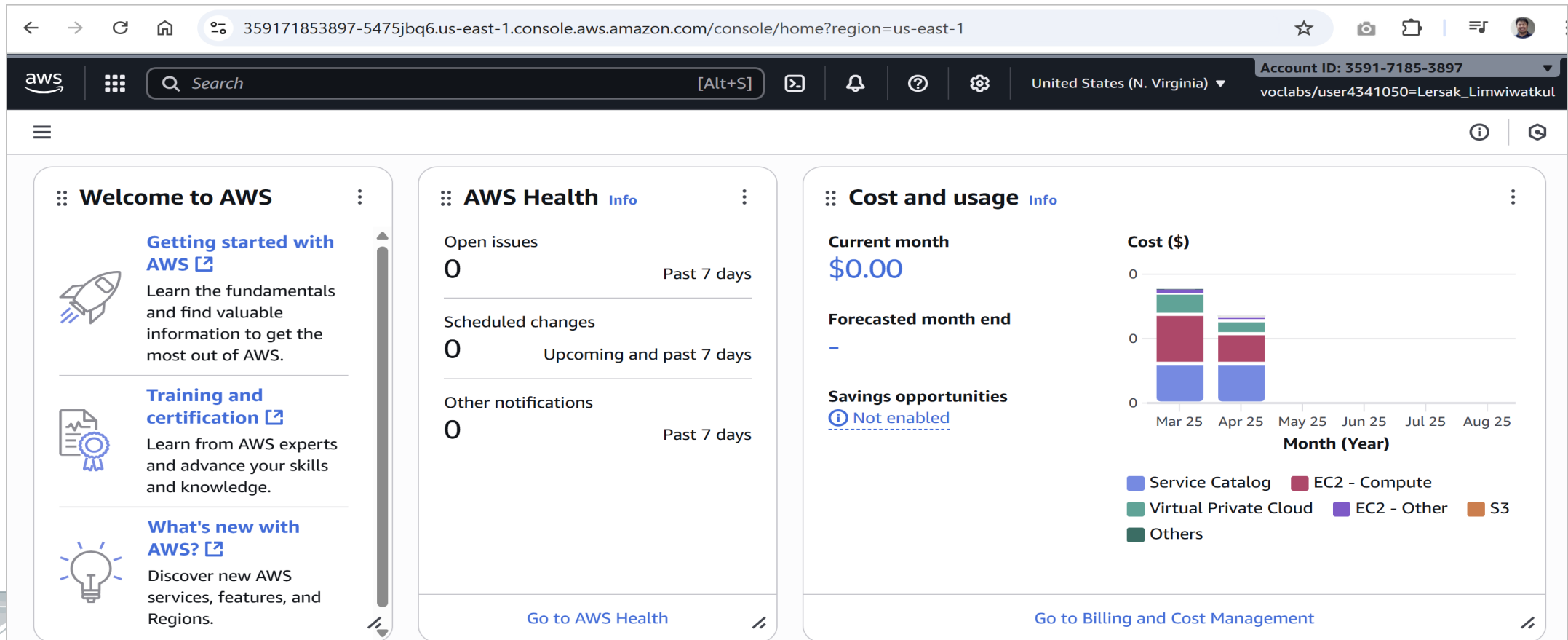


- Time elapsed for Lab
 - Start Lab – Create resources for using within Lab
 - Click again for extending Lab Time
 - End Lab – Terminate all the resources to preserve Credits
 - AWS Details – Lab Details and Access Key Information
- Wait until Lab Status **Show Green**



Access AWS Console

- Click Green AWS
- Open New Tab for AWS Console

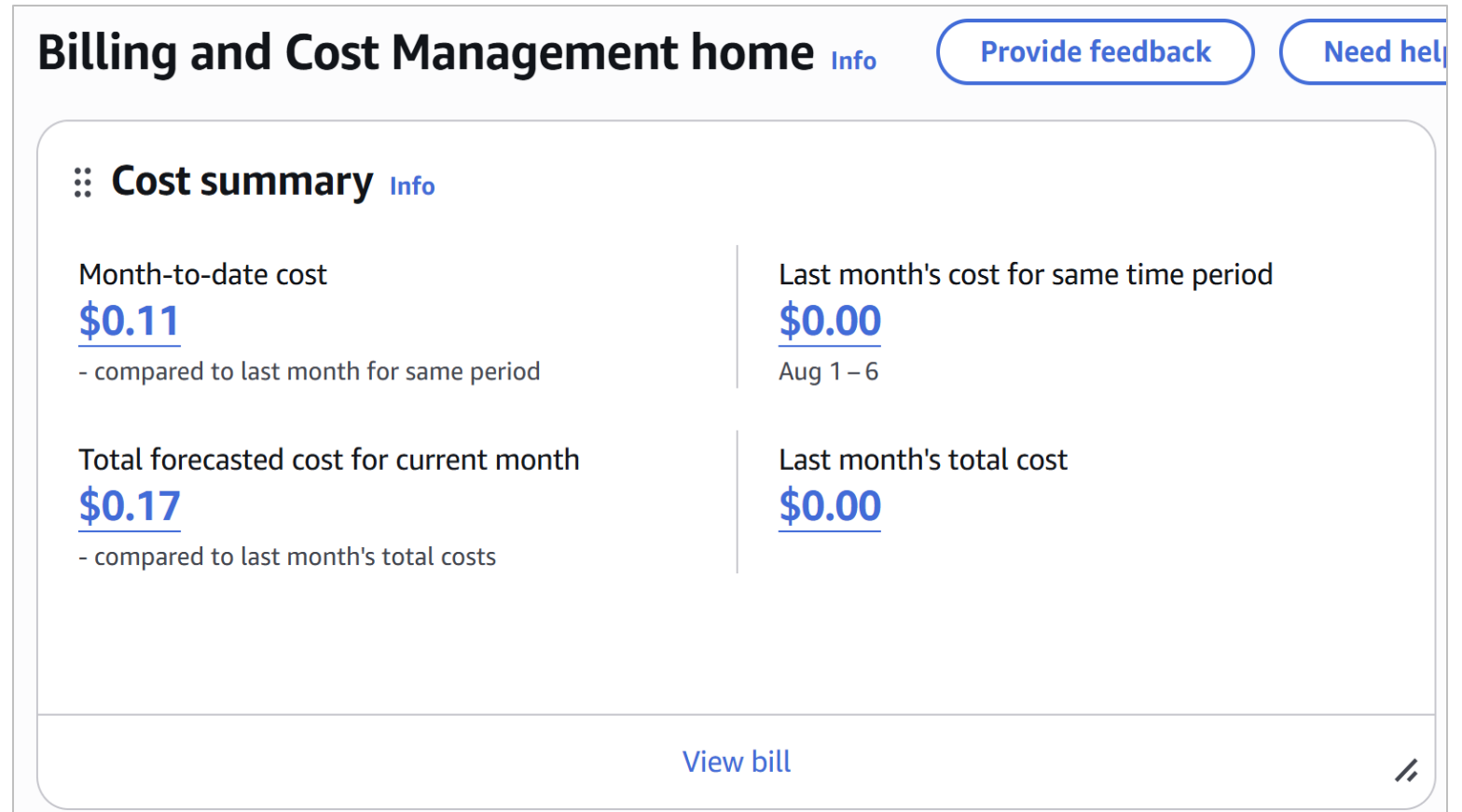
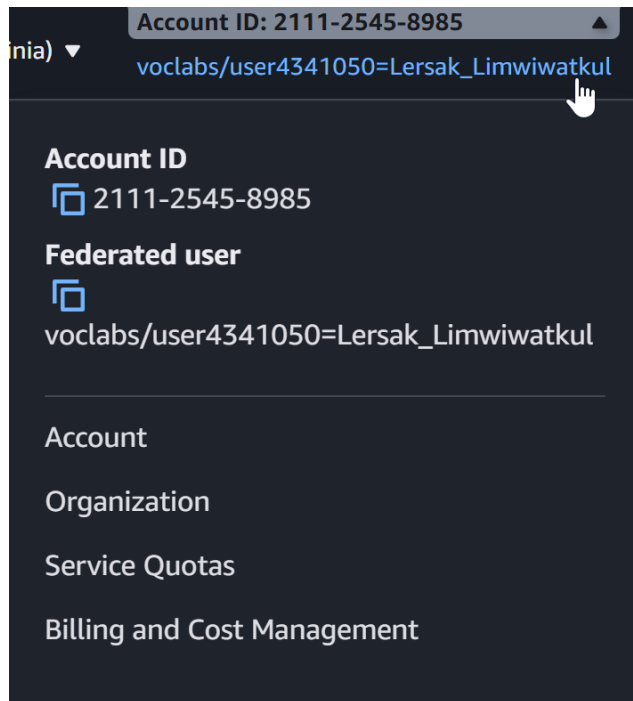


The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and various utility icons. The main content area is divided into three columns:

- Welcome to AWS:** Contains links for 'Getting started with AWS', 'Training and certification', and 'What's new with AWS?'. Each link is accompanied by an icon (rocket, certificate, and lightbulb respectively).
- AWS Health:** Displays 'Open issues' (0), 'Scheduled changes' (0), and 'Other notifications' (0). It includes a 'Go to AWS Health' link at the bottom.
- Cost and usage:** Shows 'Current month' cost as '\$0.00' and 'Forecasted month end' as '-'. It also lists 'Savings opportunities' as 'Not enabled'. A bar chart titled 'Cost (\$)' shows usage for March and April, categorized by Service Catalog, EC2 - Compute, Virtual Private Cloud, EC2 - Other, and S3. A 'Go to Billing and Cost Management' link is at the bottom.

Billings and Cost Management

- Billing and Cost Management Home



A silver golf cart is parked on a sandy beach. The cart is facing left and has a tan-colored roof. In the background, the blue ocean meets a clear blue sky with a few wispy clouds. To the right of the cart, there is a blue and white trash can. The overall scene is bright and sunny.

AWS and EC2 Console

AWS Management Console

☰

Welcome to AWS



Getting started with AWS

Learn the fundamentals and find valuable information to get the most out of AWS.



Training and certification

Learn from AWS experts and advance your skills and knowledge.



What's new with AWS?

Discover new AWS services, features, and Regions.

Go to AWS Health

☰

AWS Health

Open issues
0
Past 7 days

Scheduled changes
0
Upcoming and past 7 days

Other notifications
0
Past 7 days

Go to AWS Health

☰

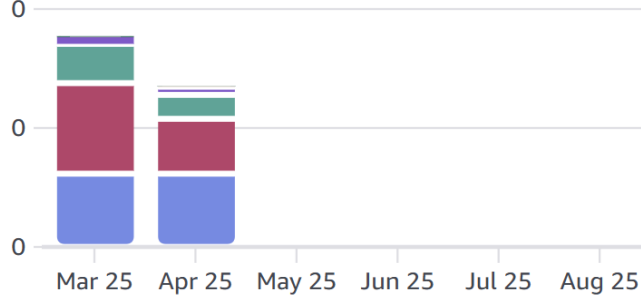
Cost and usage

Current month
\$0.00

Forecasted month end
-

Savings opportunities
Not enabled

Cost (\$)



Month (Year)	Service Catalog	EC2 - Compute	Virtual Private Cloud	EC2 - Other	S3	Others
Mar 25	100	100	50	20	0	0
Apr 25	100	100	50	20	0	0
May 25	0	0	0	0	0	0
Jun 25	0	0	0	0	0	0
Jul 25	0	0	0	0	0	0
Aug 25	0	0	0	0	0	0

Month (Year)

Service Catalog

EC2 - Compute

Virtual Private Cloud

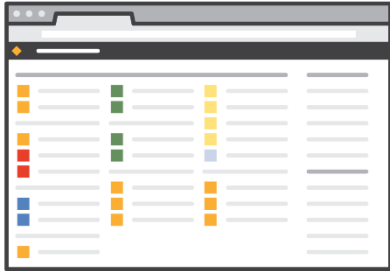
EC2 - Other

S3

Others

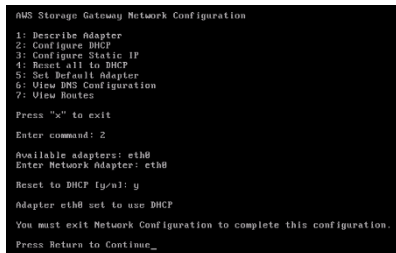
Go to Billing and Cost Management

Three ways to interact with AWS



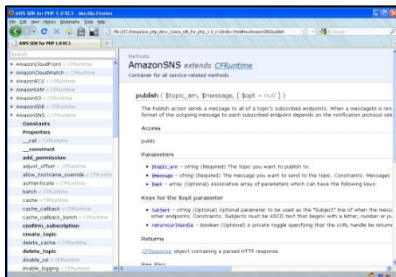
AWS Management Console

Easy-to-use graphical interface



Command Line Interface (AWS CLI)

Access to services by discrete commands or scripts

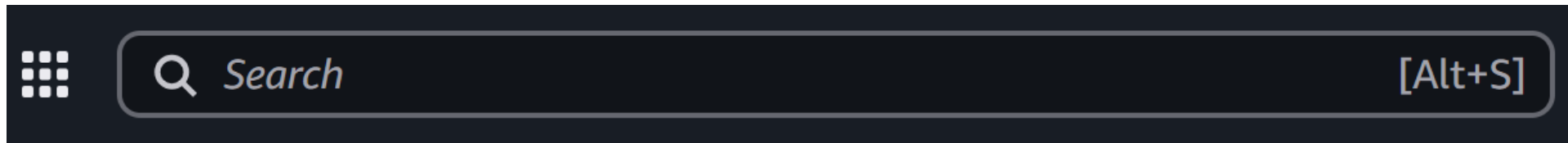


Software Development Kits (SDKs)

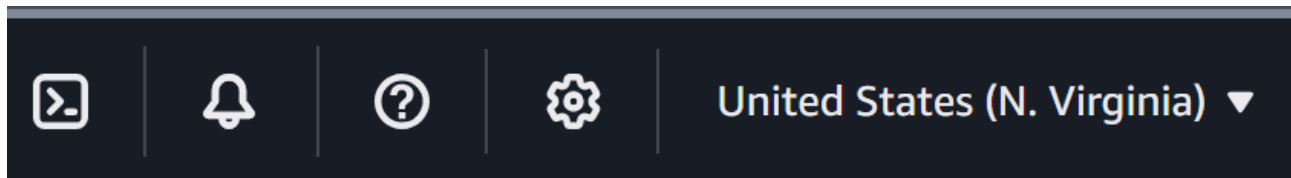
Access services directly from your code (such as Java, Python, and others)

AWS Console

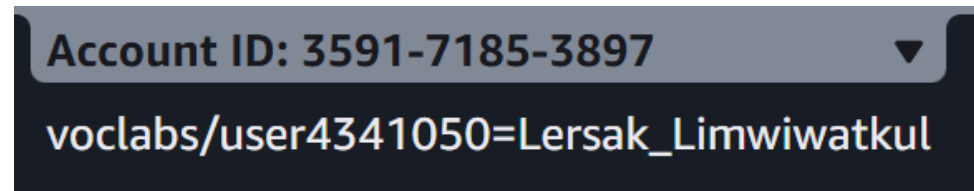
- Access all Services
- Search



- Cloud Panel

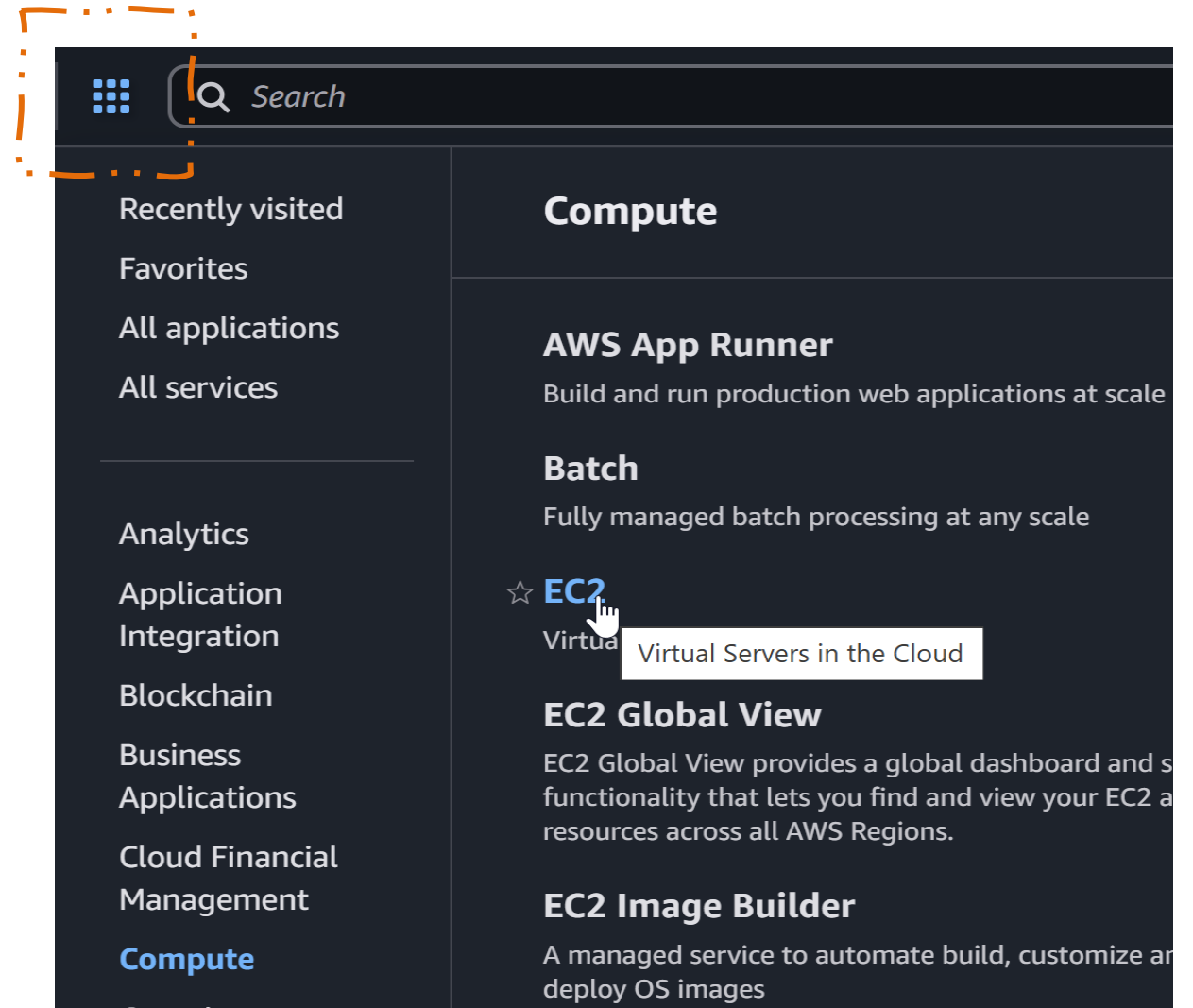


- Cloud Credential

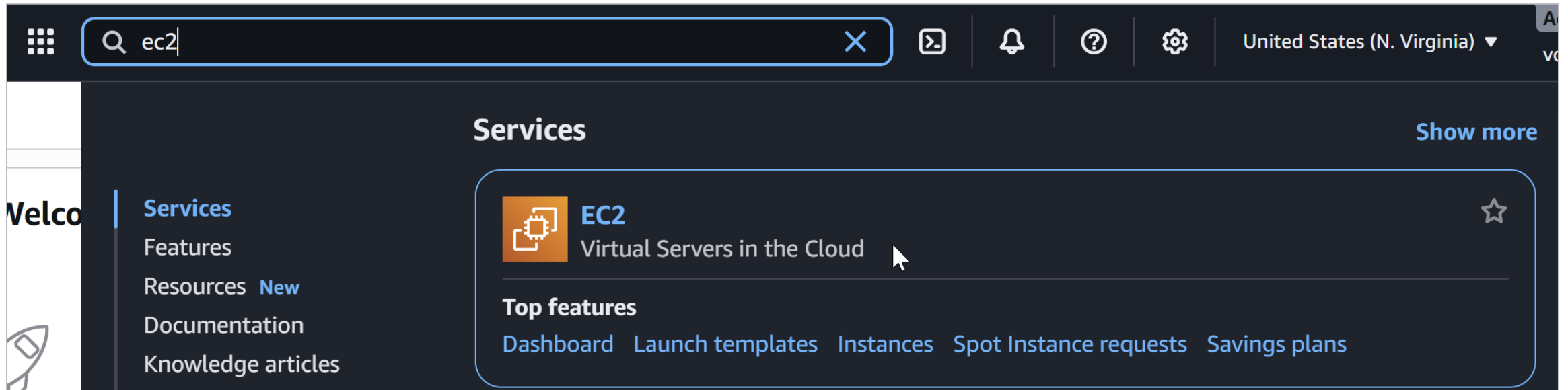


Method 1: Access AWS EC2 – Menu

- Click All Services
- Compute
- EC2
 - Virtual Servers in the Cloud



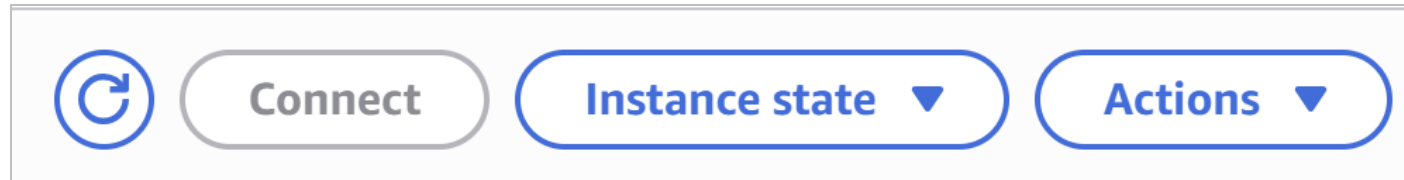
Method 2: Access AWS EC2 - Search



- Search
- EC2

EC2 Console

- Review Current Instance
- EC2 → Instances
 - Refresh
 - Instance state
 - Actions

A screenshot of the AWS Management Console showing the EC2 Instances page. The left sidebar has a menu with 'EC2' selected, and 'Instances' is highlighted with an orange dashed box. The main content area shows a table of instances with one instance listed: 'Bastion Host' with ID 'i-00b7b9e37950d21c0', in a 'Running' state, of type 't2.micro'. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, and Alarm status. Above the table are search and filter controls, and a 'Launch instances' button is on the right.

EC2 > Instances

Instances (1) Info

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	Bastion Host	i-00b7b9e37950d21c0	Running	t2.micro	2/2 checks passed	View alarms +

Instances

Instances

Instance Types

Launch Templates

A silver golf cart is parked on a sandy beach. The cart is facing left and has a tan-colored roof. In the background, the blue ocean meets a clear blue sky with a few wispy clouds. To the right of the cart, there is a blue and white trash can. The overall scene is bright and sunny.

Create our Instance

Create an Instance

- EC2 → Instances
- Launch instances

The screenshot shows the AWS Management Console interface for the EC2 service. On the left, the navigation pane is visible with 'EC2' selected, and 'Instances' is highlighted under the 'Instances' section. The main content area is titled 'Instances' and includes a 'Launch instances' button. Below this, there is a search bar and a table with columns: Name, Instance ID, Instance state, Instance type, Status check, and Availability Zone. The table is currently empty, displaying 'No instances' and 'You do not have any instances in this region'.

Create an Instance - Name

- Name and tags
- Your Student ID as Machine Name
 - <NAME><ID>

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

Create an Instance – Choose AMI

- Application and OS Images
- Amazon Machine Image (AMI)
 - Ready to use OS Images
- Select **Ubuntu 24.04 LTS**

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

🔍 Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu®

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

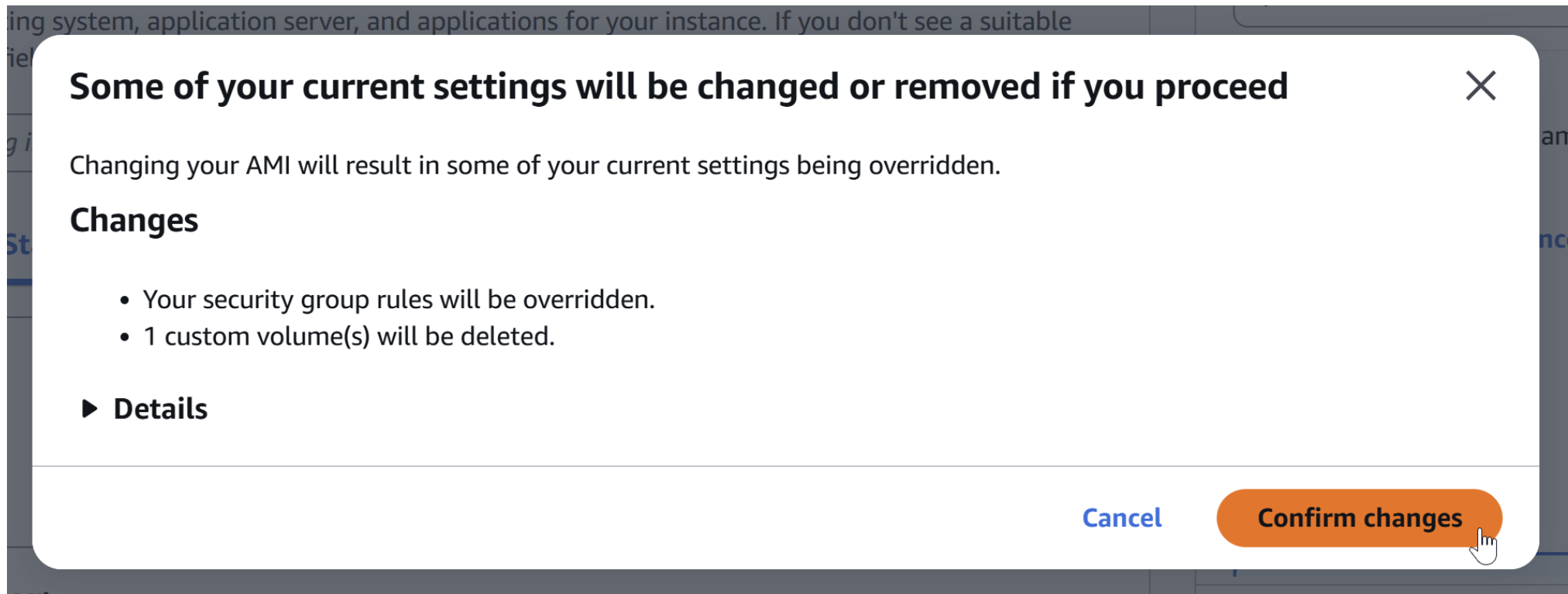
Amazon Machine Image (AMI)

Ubuntu Server **24.04 LTS** (HVM), SSD Volume Type
ami-0360c520857e3138f (64-bit (x86)) / ami-026fccd88446aa0bf (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Create an Instance – Choose AMI

- If Pop-up
- Click **Confirm changes**



Create an Instance – Select Instance type

- Instance type
- Select **t3.micro**

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Create an Instance – Select Instance type

- Select Compare instance types

Compare instance types

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instances are virtual servers that can run applications, store data, and provide networking capacity, and give you the flexibility to choose the appropriate mix of resources for your applications. Learn more about Amazon EC2 instance types.

Currently selected: t1.micro (1 vCPUs, 627 memory, EBS only)

Instance types (1/979+) [Get advice](#)

Find resources by attribute or tag

	Instance type ▼	vCPUs ▼	Architecture ▼	Memory (GiB) ▼	Storage (GB) ▼	Storage type
☒	t1.micro	1	i386, x86_64	0.612	-	-
☐	t2.nano	1	i386, x86_64	0.5	-	-
☐	t2.micro	1	i386, x86_64	1	-	-
☐	t2.small	1	i386, x86_64	2	-	-
☐	t2.medium	2	i386, x86_64	4	-	-
☐	t2.large	2	x86_64	8	-	-
☐	t2.xlarge	4	x86_64	16	-	-

Create an Instance – Select Instance type

- Instance type
 - Name
 - vCPU
 - Memory
 - Storage
 - Network Performance

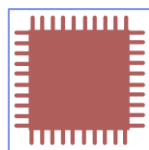
Instance Name	vCPU	Memory (GB)	Storage
t3.nano	2	0.5	EBS-Only
t3.micro	2	1	EBS-Only
t3.small	2	2	EBS-Only
t3.medium	2	4	EBS-Only
t3.large	2	8	EBS-Only
t3.xlarge	4	16	EBS-Only
t3.2xlarge	8	32	EBS-Only

Create an Instance – Instance type

- Instance Family



**General
Purpose**



**Compute
Optimized**



**Memory
Optimized**



**Accelerated
Computing**



**Storage
Optimized**

Instance Types	a1, m4, m5, t2, t3	c4, c5	r4, r5, x1, z1	f1, g3, g4, p2, p3	d2, h1, i3
Use Case	Broad	High performance	In-memory databases	Machine learning	Distributed file systems

Create an Instance – Key pair (Login)

- Select Key to Login to our Instance
- Select **vockey**

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

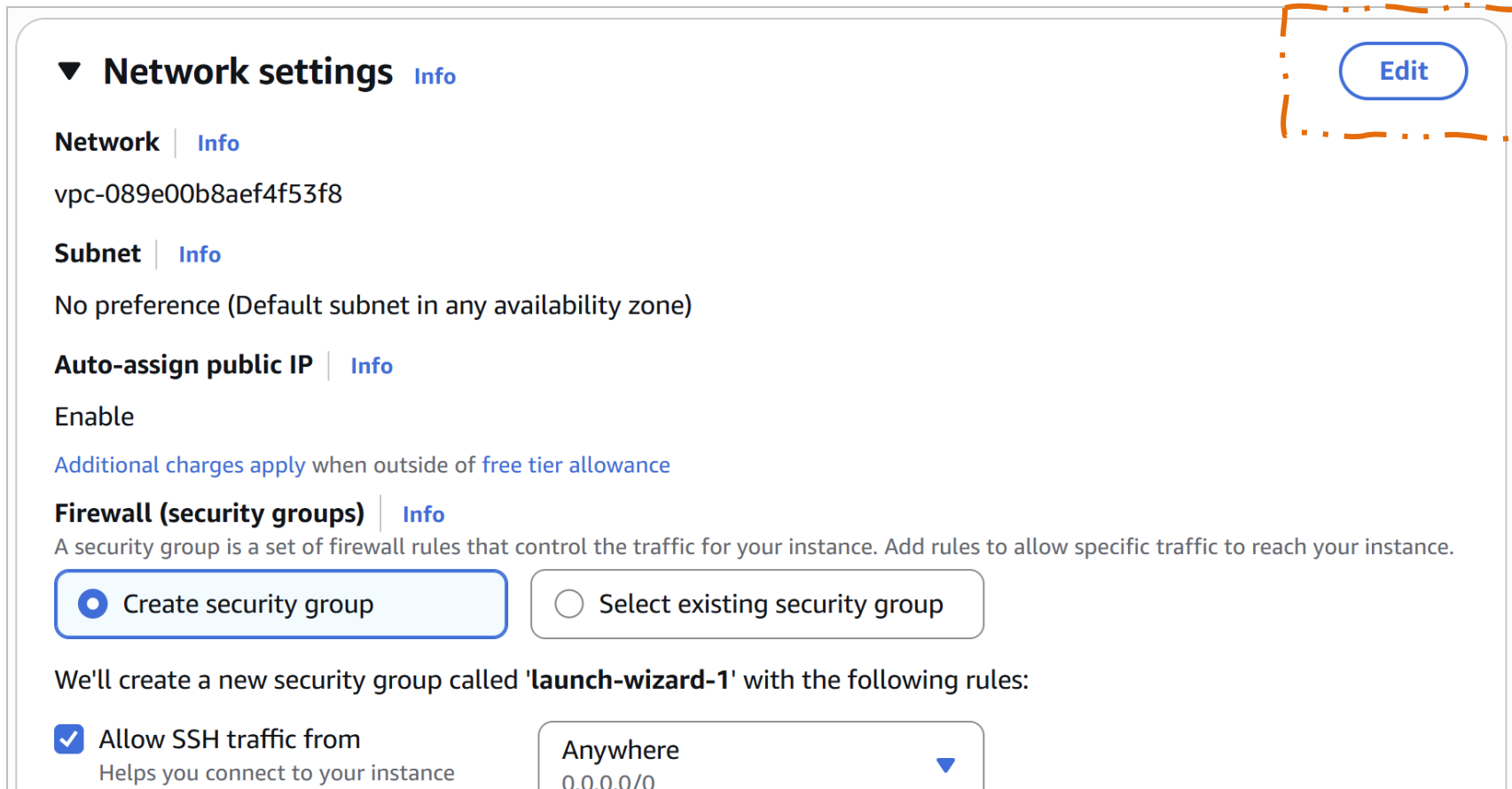
vockey



[Create new key pair](#)

Create an Instance – Network settings

- Network settings → Edits



▼ **Network settings** [Info](#)

Network | [Info](#)
vpc-089e00b8aef4f53f8

Subnet | [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)
Enable
[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) | [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called '**launch-wizard-1**' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance

Anywhere
0.0.0.0/0 ▼

Create an Instance – Network settings

- Network settings
 - VPC
 - Subnet
 - Auto-assign public IP

▼ Network settings [Info](#)

VPC - required | [Info](#)

vpc-089e00b8aef4f53f8 (default) ▼
172.31.0.0/16

Subnet | [Info](#)

subnet-031d587dea9244a5a ▼
VPC: vpc-089e00b8aef4f53f8 Owner: 211125458985
Availability Zone: us-east-1a (use1-az2) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 172.31.80.0/20

Auto-assign public IP | [Info](#)

Enable ▼

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Create an Instance – Configure storage

- Disk at 8 GiB

▼ **Configure storage** [Info](#)

Advanced

1x GiB ▼ Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

🔄 Click refresh to view backup information

↻

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

Create an Instance – Launch instance

- Summary panel of the right side
- Review Instance information
- Click **Launch instance**

▼ **Summary**

Number of instances | [Info](#)

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0360c520857e3138f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#)

Create an Instance – Launch instance

- Creating an Instance

 [EC2](#) > [Instances](#) > Launch an instance

⌵

Launching instance

Launch initiation

80%

► Details

Please wait while we launch your instance.

Do not close your browser while this is loading.

Create an Instance – Launch instance

- Successfully Launched an Instance

 [EC2](#) > [Instances](#) > Launch an instance

 **Success**
Successfully initiated launch of instance ([i-0b57d0a903748c320](#))

▼ Launch log

Initializing requests	 Succeeded
Creating security groups	 Succeeded
Creating security group rules	 Succeeded
Launch initiation	 Succeeded

Review our machine

- Click **Instances**

 [EC2](#) > **Instances** > Launch an instance

 **Success**
Successfully initiated launch of instance ([i-0b57d0a903748c320](#))

▼ Launch log

Initializing requests	 Succeeded
Creating security groups	 Succeeded
Creating security group rules	 Succeeded
Launch initiation	 Succeeded

Review our machine

- EC2 → Instances

EC2 > Instances

Instances (1) Info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set

Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	joe111111	i-0057eed8cd25d4b47	Running	t3.micro	Initializing	us-east-1c

- Select our instance

☒	joe111111	i-0b57d0a903748c320	Running	t3.micro	Initializ
-------------------------------------	-----------	---------------------	---------	----------	-----------

Review our machine

Instance summary for i-0057eed8cd25d4b47 (joe111111) [Info](#)

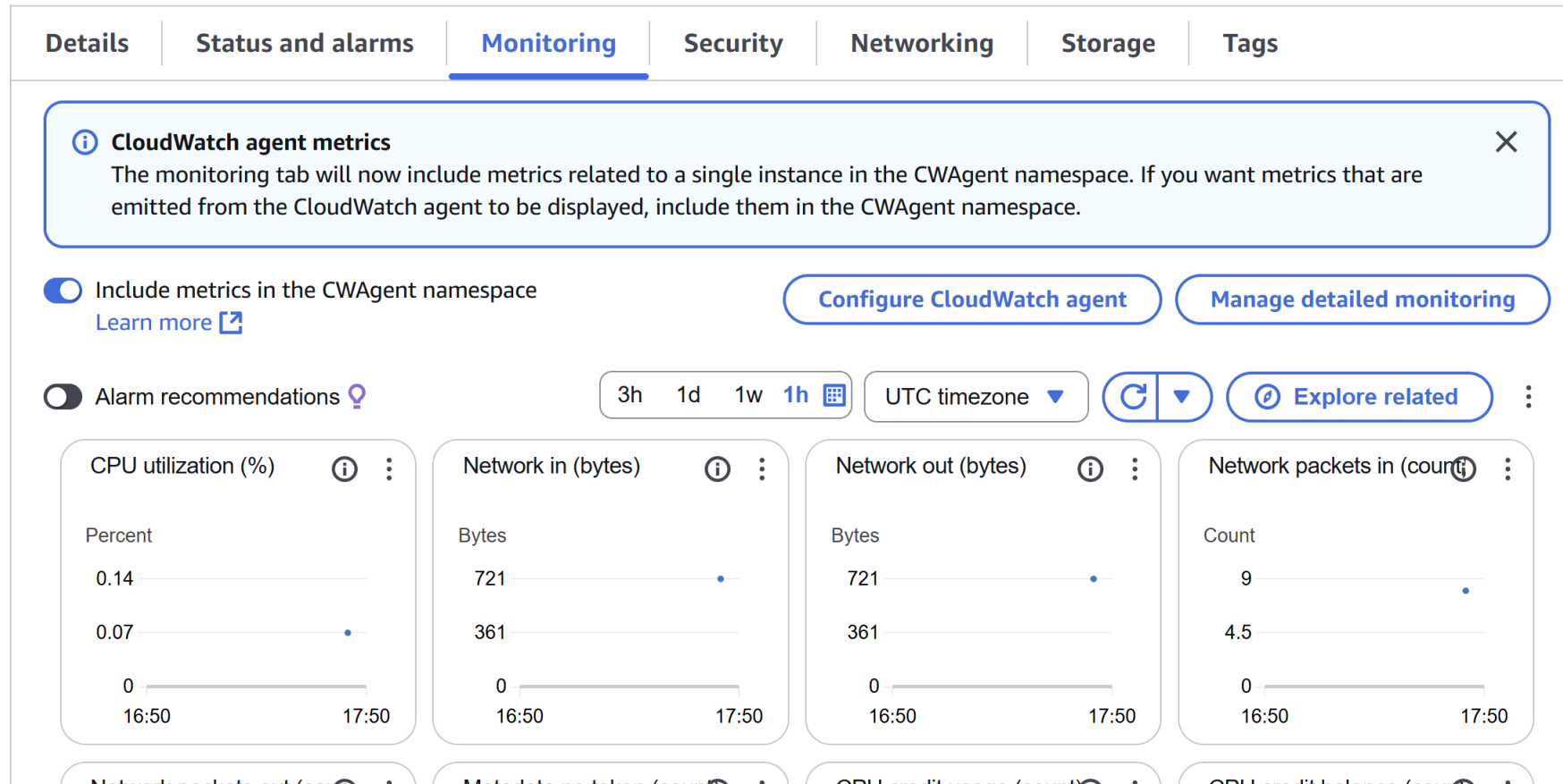
 [Connect](#) [Instance state ▼](#) [Actions ▼](#)

Updated less than a minute ago

Instance ID  i-0057eed8cd25d4b47	Public IPv4 address  98.87.10.197 open address ↗	Private IPv4 addresses  172.31.35.4
IPv6 address —	Instance state  Running	Public DNS  ec2-98-87-10-197.compute-1.amazonaws.com open address ↗
Hostname type IP name: ip-172-31-35-4.ec2.internal	Private IP DNS name (IPv4 only)  ip-172-31-35-4.ec2.internal	Elastic IP addresses —
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding  Opt-in to AWS Compute Optimizer for recommendations. Learn more ↗
Auto-assigned IP address  98.87.10.197 [Public IP]	VPC ID  vpc-089e00b8aef4f53f8 ↗	

Review our machine – Monitoring Panel

- Scroll down and select **Monitoring**



Manage Security

Manage Security Rule = Firewall

- Select Security

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role

–

Owner ID

 359171853897

Launch time

Mon Sep 01 2025 00:46:33 GMT+0700 (Indochina Time)

Security groups

 sg-0058013136c8f56d3 (launch-wizard-1)

▼ Inbound rules

 Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Source
–	sgr-0091ec6b5848771ce	22	TCP	0.0.0.0/0

Manage Security Rule = Firewall

- Click Security groups

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role
–

Owner ID
 359171853897

Launch time
Mon Sep 01 2025 00:46:33 GMT+0700
(Indochina Time)

Security groups

 sg-0058013136c8f56d3 (launch-wizard-1)

▼ Inbound rules

 Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Source
–	sgr-0091ec6b5848771ce	22	TCP	0.0.0.0/0

Manage Security Rule = Firewall

- Actions → Edit inbound rules

sg-0058013136c8f56d3 - launch-wizard-1

Actions ▲

- Edit inbound rules
- Edit outbound rules
- Share security group
- Manage tags
- Copy to new security group
- Delete security groups

Details

Security group name launch-wizard-1	Security group ID sg-0058013136c8f56d3	Description launch-wizard-1 created 2025-08-31T17:24:44.767Z
Owner 359171853897	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry

[Inbound rules](#) | [Outbound rules](#) | [Sharing - new](#) | [VPC associations - new](#) | [Tags](#)

Inbound rules (1)

[Manage tags](#)[Edit inbound rules](#)

Manage Security Rule = Firewall

- Click **Add rule** → **Add Rule** → **Save rules**

Edit inbound rule

Inbound rules control the incoming traffic to your instance.

Inbound rules

Security group rule ID

sgr-0091ec6b5848771c

Add rule

Inbound rules

Security group rule ID

Security group rule ID	Type	Protocol	Port range	Source
sgr-0f8284f7371cac0e3	SSH	TCP	22	Cus...
-	All ICMP - IPv4	ICMP	All	An...

0.0.0.0/0

0.0.0.0/0

0.0.0.0/0

Cancel Preview changes Save rules

Manage Security Rule = Firewall

Inbound rules Info						
Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description
sgr-0b7ceabbc313a6b5e	HTTP ▼	TCP	80	An... ▼	0.0.0.0/0 ✕	
sgr-0f8284f7371cac0e3	SSH ▼	TCP	22	An... ▼	0.0.0.0/0 ✕	
sgr-088440de4045c7cbb	All ICMP - IPv4 ▼	ICMP	All	An... ▼	0.0.0.0/0 ✕	

- SSH from 0.0.0.0/0
- ICMP from 0.0.0.0/0
- HTTP from 0.0.0.0/0

Manage Security Rule = Firewall

- Review Inbound rules

Inbound rules (3)								Manage tags	Edit inbound rules
Search							< 1 > 		
☐	Name	Security group rule ID	IP version	Type	Protocol				
☐	-	sgr-0a149d40836cf26eb	IPv4	HTTP	TCP				
☐	-	sgr-0378f9ceede88bc14	IPv4	All ICMP - IPv4	ICMP				
☐	-	sgr-0091ec6b5848771ce	IPv4	SSH	TCP				

- HTTP
- All ICMP
- SSH

A silver golf cart is parked on a sandy beach. The cart has a tan roof and two seats. In the background, the ocean stretches to the horizon under a clear blue sky with a few wispy clouds. To the right of the cart, there is a blue and white trash can. The text "Access your Machine" is overlaid in white on the lower left side of the image.

Access your Machine

Retrieve IP Address

- EC2 → Instances

EC2 > Instances

Instances (1) Info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set

Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	joe111111	i-0057eed8cd25d4b47	Running	t3.micro	Initializing	us-east-1c

- Select our instance

☒	joe111111	i-0b57d0a903748c320	Running	t3.micro	Initializing
-------------------------------------	-----------	---------------------	---------	----------	--------------

Retrieve IP Address

Instance summary for i-0057eed8cd25d4b47 (joe111111) [Info](#)

 [Connect](#) [Instance state ▼](#) [Actions ▼](#)

Updated less than a minute ago

Instance ID  i-0057eed8cd25d4b47	Public IPv4 address  98.87.10.197 open address ↗	Private IPv4 addresses  172.31.35.4
IPv6 address —	Instance state  Running	Public DNS  ec2-98-87-10-197.compute-1.amazonaws.com open address ↗
Hostname type IP name: ip-172-31-35-4.ec2.internal	Private IP DNS name (IPv4 only)  ip-172-31-35-4.ec2.internal	Elastic IP addresses —
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding  Opt-in to AWS Compute Optimizer for recommendations. Learn more ↗
Auto-assigned IP address  98.87.10.197 [Public IP]	VPC ID  vpc-089e00b8aef4f53f8 ↗	

Ping to our Machine

- Start → cmd

ping <ip>

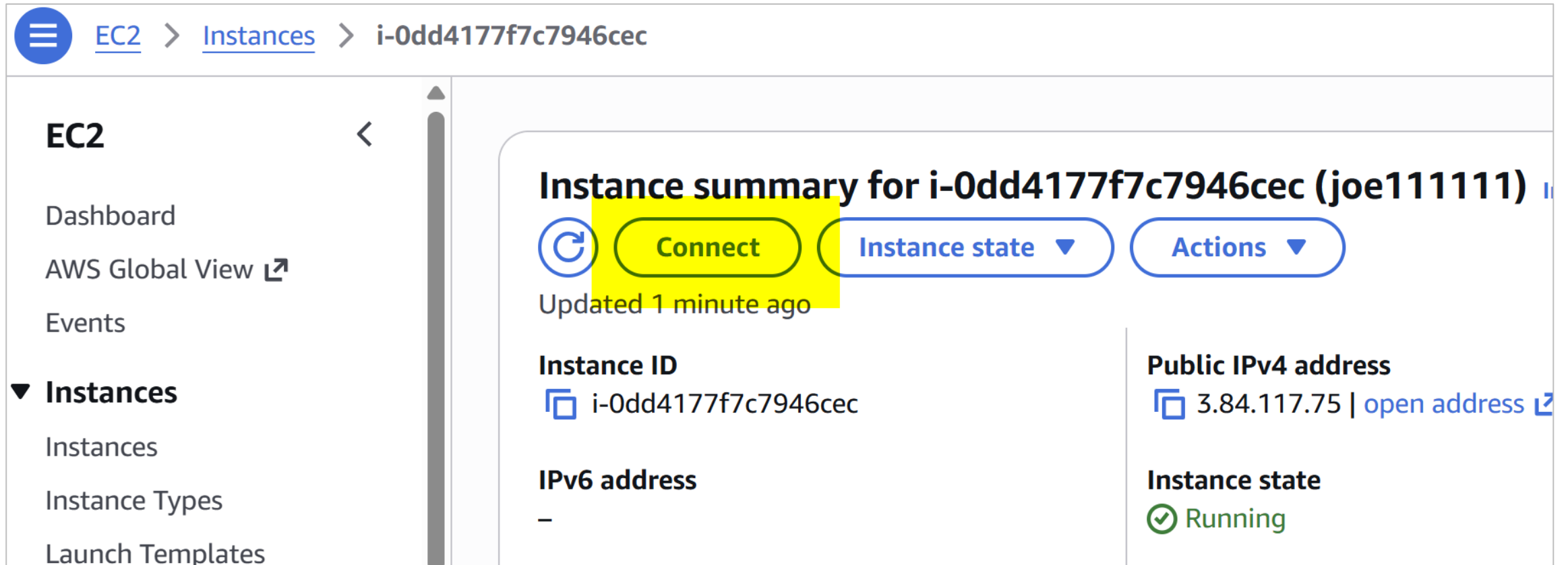
```
C:\Users\lorsa>
C:\Users\lorsa>ping 3.84.117.75

Pinging 3.84.117.75 with 32 bytes of data:
Reply from 3.84.117.75: bytes=32 time=337ms TTL=48
Reply from 3.84.117.75: bytes=32 time=265ms TTL=48
Reply from 3.84.117.75: bytes=32 time=276ms TTL=48
Reply from 3.84.117.75: bytes=32 time=272ms TTL=48

Ping statistics for 3.84.117.75:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

Access to our Machine

- EC2 → Instances → Detail → Click **Connect**



The screenshot shows the AWS Management Console interface for an EC2 instance. The breadcrumb navigation at the top reads: **EC2** > **Instances** > **i-0dd4177f7c7946cec**. On the left sidebar, under the **EC2** section, the **Instances** link is expanded, showing a list of instance-related options. The main content area displays the **Instance summary for i-0dd4177f7c7946cec (joe111111)**. A yellow highlight is placed over the **Connect** button, which is located next to a refresh icon and the text 'Updated 1 minute ago'. Other buttons visible include **Instance state** (with a dropdown arrow) and **Actions** (with a dropdown arrow). Below the summary, the **Instance ID** is shown as **i-0dd4177f7c7946cec**. The **IPv6 address** is shown as **—**. The **Public IPv4 address** is shown as **3.84.117.75** with a link to **open address**. The **Instance state** is shown as **Running** with a green checkmark icon.

EC2 > Instances > i-0dd4177f7c7946cec

EC2

Dashboard

AWS Global View ↗

Events

▼ **Instances**

Instances

Instance Types

Launch Templates

Instance summary for i-0dd4177f7c7946cec (joe111111)

⌂ **Connect** Instance state ▼ Actions ▼

Updated 1 minute ago

Instance ID

⌂ i-0dd4177f7c7946cec

IPv6 address

—

Public IPv4 address

⌂ 3.84.117.75 | open address ↗

Instance state

✓ Running

Access to our Machine

- Review Status = 3/3 passed

Connect to Linux instance [Info](#)

Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser

Instance info

Instance

 i-odd4177f7c7946cec (joe111111)

 State: Running

 Status: 3/3 checks passed

VPC 

 [vpc-089e00b8aef4f53f8](#)

In web browser

In SSH client

Access to our Machine

- Scroll down
- Click **Connect**

☒ IPv4 address ( 3.84.117.75)

☐ IPv6 address

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user. Read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

 ubuntu 

Port 22 (SSH) is open to all IPv4 addresses

Port 22 (SSH) is currently open to all IPv4 addresses, indicated by 0.0.0.0/0 in the inbound rule in [your security group](#). For increased security, consider restricting access to only the EC2 Instance Connect service IP addresses for your Region: 18.206.107.24/29. [Learn more](#)

Cancel

 **Connect assist**

Connect 

Access to our Machine

- New tab
- Console
- Prompt = \$

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&ins

aws [Alt+S] Search United States

* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Mon Aug 24 17:55:22 UTC 2026

System load: 0.02 Temperature: -273.1 C
Usage of /: 26.6% of 6.71GB Processes: 115
Memory usage: 25% Users logged in: 0
Swap usage: 0% IPv4 address for ens5: 172.31.83.189

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-83-189:~$
```


Review your Linux Machine

- Login user

```
$ id
```

- System information

```
$ uname -a
```

- Memory information

```
$ free -h
```

- Disk information

```
$ df -h
```

Review your Linux Machine

```
ubuntu@ip-172-31-83-189:~$  
ubuntu@ip-172-31-83-189:~$ id  
uid=1000(ubuntu) gid=1000(ubuntu) groups=1000(ubuntu),4(adm),24(cdrom),27(sudo),30(dialout),44(mgmt),46(network)ubuntu@ip-172-31-83-189:~$  
ubuntu@ip-172-31-83-189:~$ uname -a  
Linux ip-172-31-83-189 6.17.0-1017-aws #17~24.04.1-Ubuntu SMP Tue May 26 21:30:32 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux  
ubuntu@ip-172-31-83-189:~$ free -h  
               total        used        free      shared  buff/cache   available  
Mem:           911Mi       351Mi       296Mi         2.7Mi        420Mi        559Mi  
Swap:              0B           0B           0B  
ubuntu@ip-172-31-83-189:~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/root        6.8G  1.9G  4.9G  28% /  
tmpfs            456M   0  456M   0% /dev/shm  
tmpfs            183M  872K  182M   1% /run  
tmpfs            5.0M   0   5.0M   0% /run/lock  
efivarfs         128K   3.1K  120K   3% /sys/firmware/efi/efivars  
/dev/nvme0n1p16  881M   94M  726M  12% /boot  
/dev/nvme0n1p15  105M   6.2M   99M   6% /boot/efi  
tmpfs            92M   12K   92M   1% /run/user/1000  
ubuntu@ip-172-31-83-189:~$
```

Setup Web Server

Setup Web Server

- Update Linux Machine

```
$ sudo apt update && sudo apt upgrade -y
```

```
ubuntu@ip-172-31-83-189:~$  
ubuntu@ip-172-31-83-189:~$ sudo apt update && sudo apt upgrade -y  
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease  
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]  
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [969 kB]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [206 kB]  
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]  
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]  
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]  
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]  
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]  
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]  
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]  
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1229 kB]  
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [286 kB]
```


Setup Web Server

- Install Nginx Web Server

```
$ sudo apt install nginx -y
```

```
ubuntu@ip-172-31-83-189:~$  
ubuntu@ip-172-31-83-189:~$ sudo apt install nginx -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  nginx-common  
Suggested packages:  
  fcgiwrap nginx-doc ssl-cert  
The following NEW packages will be installed:  
  nginx nginx-common  
0 upgraded, 2 newly installed, 0 to remove and 1 not upgraded.  
Need to get 568 kB of archives.  
After this operation, 1602 kB of additional disk space will be  
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-upda  
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-upda  
Fetched 568 kB in 0s (8110 kB/s)  
Preconfiguring packages ...  
Selecting previously unselected package nginx-common.
```

Setup Web Server

- Review Status

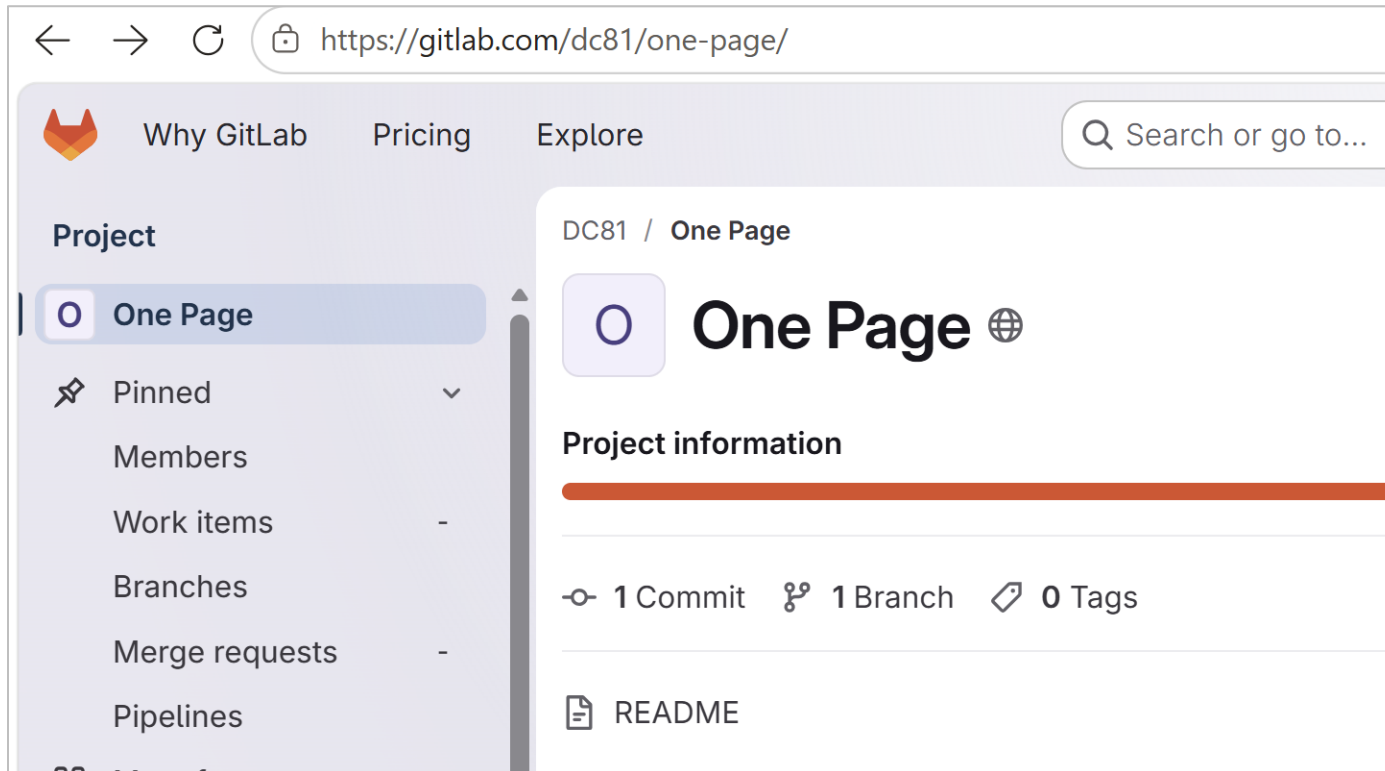
```
$ sudo systemctl status nginx
```

```
no VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-83-189:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-24 18:13:53 UTC; 1min 5s ago
     Docs: man:nginx(8)
  Process: 25733 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on;
  Process: 25735 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=ex
Main PID: 25764 (nginx)
   Tasks: 3 (limit: 1013)
  Memory: 2.4M (peak: 5.2M)
     CPU: 28ms
   CGroup: /system.slice/nginx.service
           └─25764 "nginx: master process /usr/sbin/nginx -g daemon on; master_proce
              └─25767 "nginx: worker process"
                 └─25768 "nginx: worker process"
```

Enable Web Content

- Access One Page Content

<https://gitlab.com/dc81/one-page/>



Enable Web Content

- Download One Page

```
$ wget https://gitlab.com/dc81/one-page/-/raw/main/index.html
```

```
ubuntu@ip-172-31-83-189:~$  
ubuntu@ip-172-31-83-189:~$ wget https://gitlab.com/dc81/one-page/-/raw/main/index.html  
--2026-08-24 18:29:48-- https://gitlab.com/dc81/one-page/-/raw/main/index.html  
Resolving gitlab.com (gitlab.com)... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9  
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 2403 (2.3K) [text/plain]  
Saving to: 'index.html'  
  
index.html          100% [=====]  
  
2026-08-24 18:29:48 (30.7 MB/s) - 'index.html' saved [2403/2403]  
  
ubuntu@ip-172-31-83-189:~$ █
```

Enable Web Content

- Copy to Web Content

```
$ sudo cp index.html /var/www/html/index.html
```

```
$ ls -l /var/www/html/index.html
```

```
ubuntu@ip-172-31-83-189:~$ sudo cp index.html /var/www/html/index.html
ubuntu@ip-172-31-83-189:~$ ls -l /var/www/html/index.html
-rw-r--r-- 1 root root 2403 Aug 24 18:30 /var/www/html/index.html
ubuntu@ip-172-31-83-189:~$
```

Access to your Web Server

- EC2 → Instances → Instance Summary
- Public IPv4 address → Click Copy

Instance summary for i-0dd4177f7c7946cec (joe111111) [Info](#)

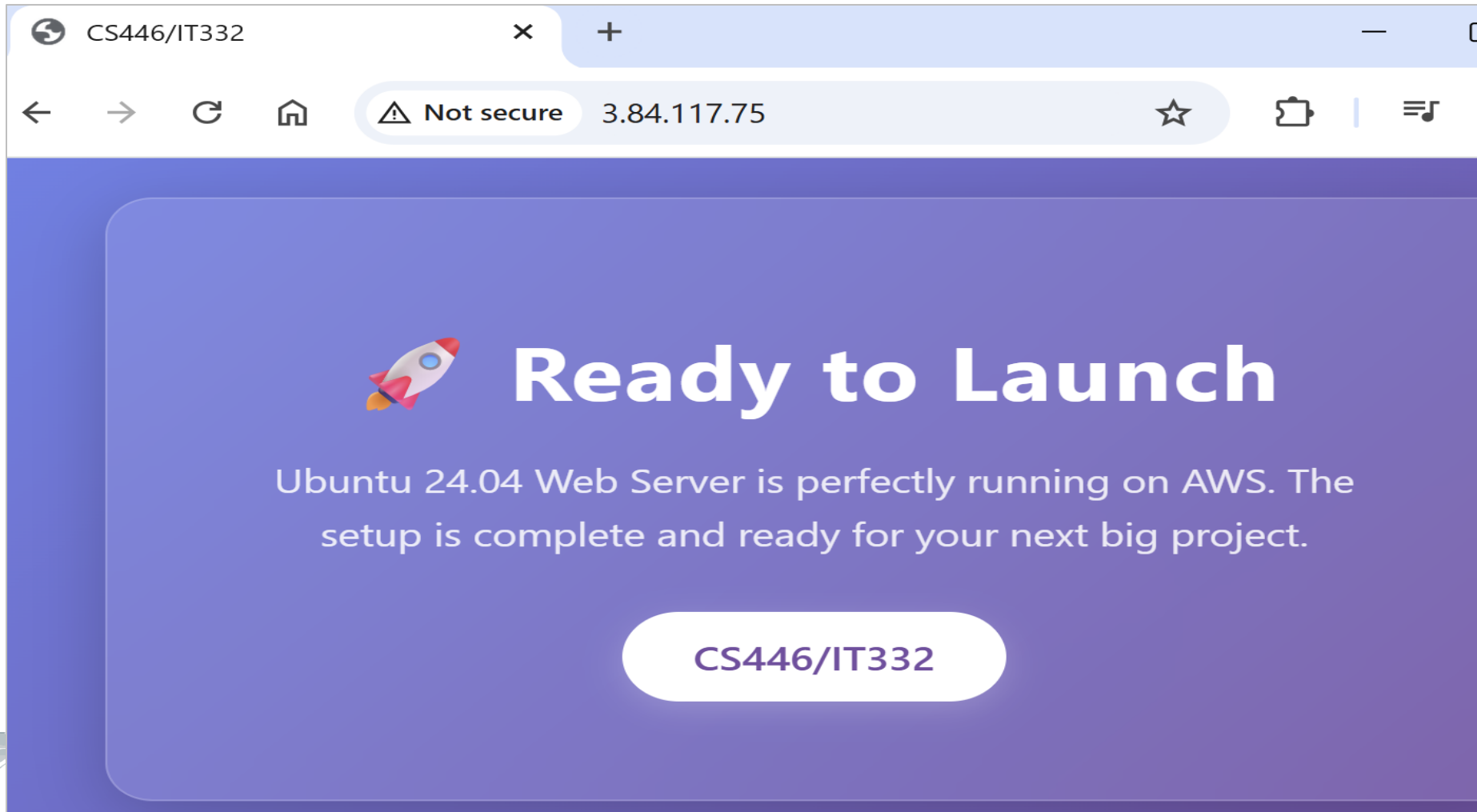
[Connect](#) [Instance state ▼](#) [Actions ▼](#)

Updated less than a minute ago

Instance ID i-0dd4177f7c7946cec	Public IPv4 address 3.84.117.75 open address ↗ Copy public IPv4 address to clipboard
IPv6 address —	Ins Running

Access to your Web Server

- Open Browser and type URL → `http://<IP>:80`

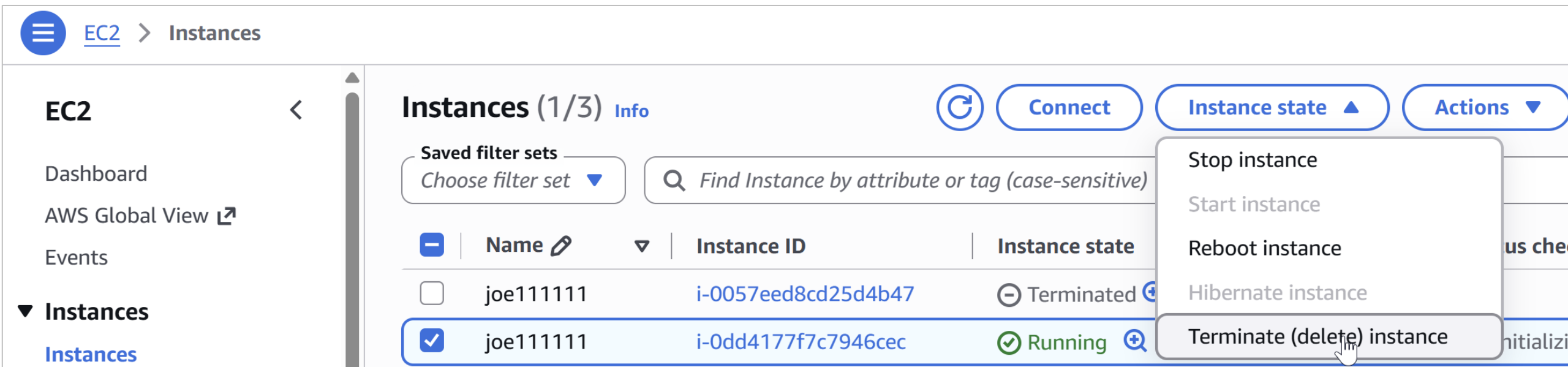


Clear Lab



Terminate EC2 Instances

- EC2 → Instances → Instances
- Instance state → Terminate (delete) instance



The screenshot displays the AWS Management Console interface for the EC2 service. The left-hand navigation pane shows the 'EC2' section with a sub-menu 'Instances' highlighted. The main content area is titled 'Instances (1/3)' and includes a search bar and a table of instances. The table has columns for 'Name', 'Instance ID', and 'Instance state'. Two instances are listed: 'joe111111' with ID 'i-0057eed8cd25d4b47' in a 'Terminated' state, and 'joe111111' with ID 'i-0dd4177f7c7946cec' in a 'Running' state. The 'Running' instance is selected. An 'Instance state' dropdown menu is open, showing options: 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. The 'Terminate (delete) instance' option is highlighted by the mouse cursor.

	Name	Instance ID	Instance state
☐	joe111111	i-0057eed8cd25d4b47	Terminated
☒	joe111111	i-0dd4177f7c7946cec	Running

Terminate EC2 Instances

- Click **Terminate (delete)**

×

Terminate (delete) instance

⚠

On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID	Termination protection
 i-0dd4177f7c7946cec (joe111111)	 Disabled

To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

Skip OS shutdown

This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.

☐ Skip OS shutdown

Cancel

Terminate (delete)

Terminate EC2 Instances

- Wait until all your instances have been terminated

Instances (3) Info							Connect	Instance state ▼	Actions ▼
Saved filter sets		Find Instance by attribute or tag (case-sensitive)							
Choose filter set ▼									
☐	Name 	▼	Instance ID	Instance state ▼	Instance type ▼	Status			
☐	joe111111		i-0057eed8cd25d4b47	⊖ Terminated  	t3.micro	–			
☐	joe111111		i-0dd4177f7c7946cec	⊖ Terminated  	t3.micro	–			
☐	joe111111		i-041ee237865beff5f	⊖ Terminated  	t3.micro	–			

AWS Pricing Model

- 3 Fundamental drivers of AWS cost

Compute

- Charged per hour/second*
- Varies by instance type

*Linux only

Storage

- Charged typically per GB

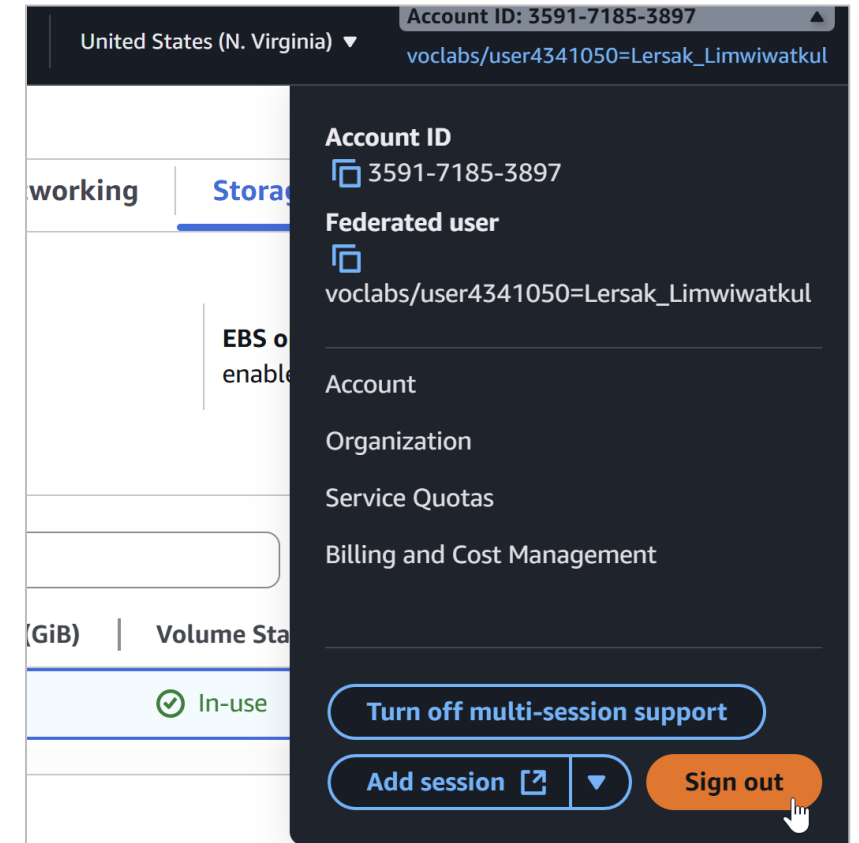
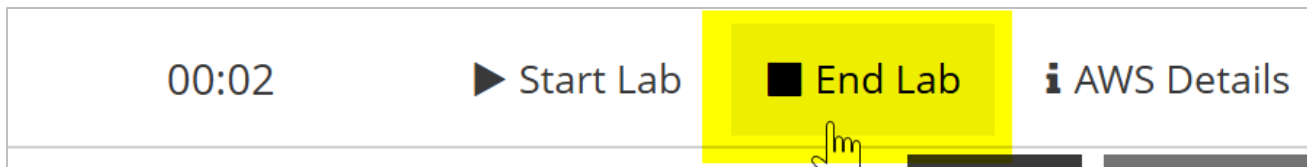
Data transfer

- Outbound is aggregated and charged
- Inbound has no charge (with some exceptions)
- Charged typically per GB

Sign out and End Lab

- Sign out from AWS Console

- Click End Lab



Always End Lab

- Click **End Lab**



- Wait until Lab Status
Become Red

